



**SYDNEY
BOYS
HIGH
SCHOOL**

NESA Number:

--	--	--	--	--	--	--	--	--

Name:

Maths Class: Circle below
A B 1 2 P L U S

2023

YEAR 12

TERM 2

HSC Task #3

Mathematics Extension 1

General Instructions

- Working Time – 65 minutes
- Write using black pen
- NESA approved calculators may be used
- Marks may **NOT** be awarded for messy or badly arranged work
- Unless otherwise stated, all answers should be left in simplified, exact form
- In Questions 11–18, show ALL relevant mathematical reasoning and/or calculations

Total Marks: Section I – 10 marks (Personal Reference sheet)

Leave your Personal Reference Sheet this booklet.

Section II – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section III – 30 marks (pages 6–8)

- Attempt Questions 11–18
- Allow about 45 minutes for this section

Examiner:

PSP

Marks	
Section I	/10
Section II	/10
Section III	/30
Total	/50

Section II

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Let $f(x) = \operatorname{cosec} x$.

The graph of f is transformed by

- a vertical dilation by a factor of 3, followed by
- a translation of 1 unit horizontally to the right, followed by
- a horizontal dilation by a factor of $\frac{1}{2}$.

Which of the following functions is the transformed graph?

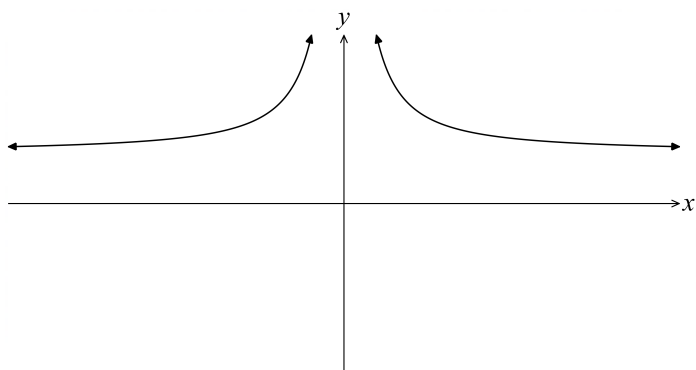
- A. $g(x) = 3\operatorname{cosec}(2x - 1)$ B. $g(x) = 3\operatorname{cosec}\left(\frac{x}{2} - 1\right)$
- C. $g(x) = 3\operatorname{cosec}(2(x - 1))$ D. $g(x) = 3\operatorname{cosec}\left(\frac{x - 1}{2}\right)$

2 Which of the following is the inverse function of $f(x) = 2x^{\frac{1}{4}}$?

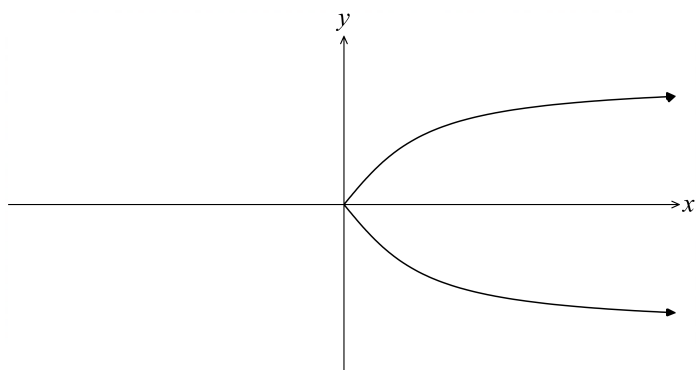
- A. $f^{-1}(x) = \frac{x^{\frac{1}{4}}}{16}$
- B. $f^{-1}(x) = \frac{x^4}{16}$
- C. $f^{-1}(x) = \frac{x^4}{8}$
- D. $f^{-1}(x) = \frac{x^4}{2}$

3 Which of the following sketches best represents $|y| = \arctan x$?

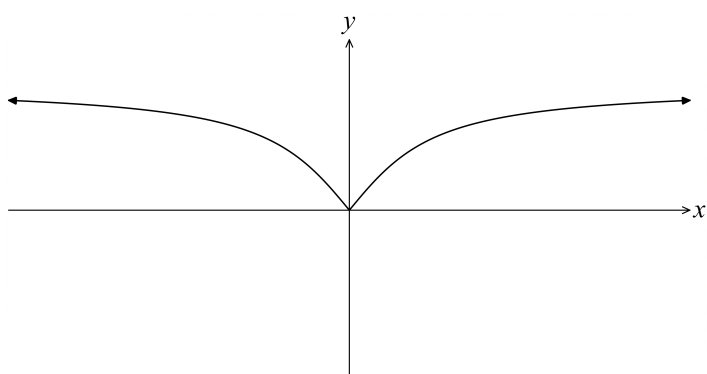
A.



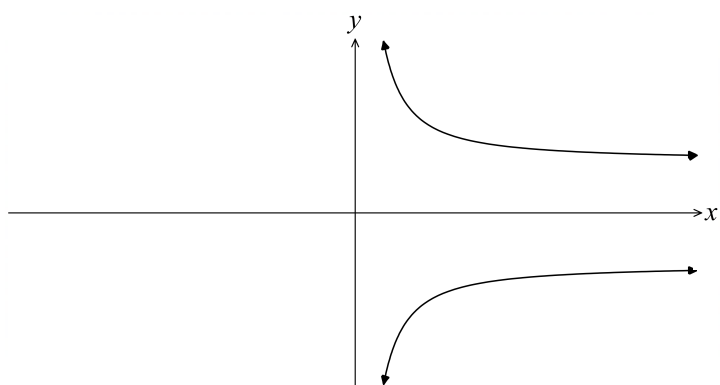
B.



C.



D.



4 Which of the following is equivalent to $\int_0^{\frac{\pi}{6}} \sin^5 x \cos x \, dx$, after applying the substitution $u = \sin x$?

A. $\int_0^{\frac{\pi}{6}} u^5 \, du$

B. $\int_0^{\frac{1}{2}} u^5 \, du$

C. $-\int_0^{\frac{\pi}{6}} u^5 \, du$

D. $-\int_0^{\frac{1}{2}} u^5 \, du$

5 Given that $f(x) = e^x - 1$ and $y = f^{-1}(x)$, which of the following is an expression for $\frac{dy}{dx}$?

A. $\frac{1}{e^x - 1}$

B. $\frac{1}{x + 1}$

C. $\ln x$

D. $\ln(x + 1)$

6 What is the value of $\arcsin(\cos \alpha)$, where $\pi < \alpha < \frac{3\pi}{2}$?

A. $\frac{\pi}{2} - \alpha$

B. $\frac{3\pi}{2} + \alpha$

C. $\alpha - \frac{3\pi}{2}$

D. $\frac{3\pi}{2} - \alpha$

7 If $y = \sin^{-1}\left(\frac{a}{x}\right)$ then which of the following is $\frac{dy}{dx}$, where a is a constant?

A. $-\frac{a}{x^2\sqrt{x^2 - a^2}}$

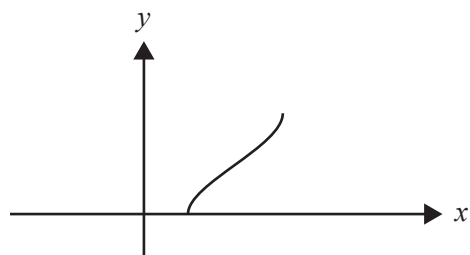
B. $-\frac{a}{x^2\sqrt{a^2 - x^2}}$

C. $-\frac{a}{|x|\sqrt{x^2 - a^2}}$

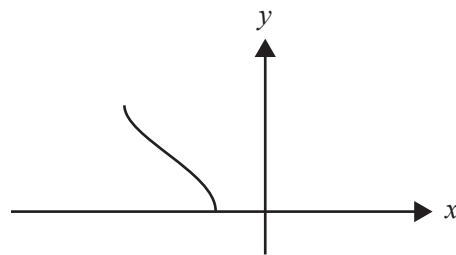
D. $-\frac{a}{|x|\sqrt{a^2 - x^2}}$

8 Which of the following could be the graph of $y = \arccos(2 - bx)$, where $b > 0$?

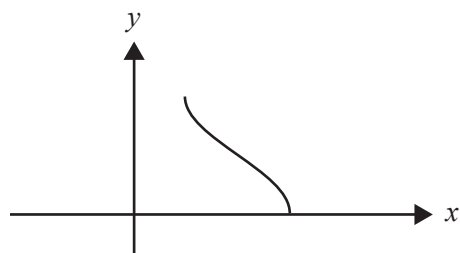
A.



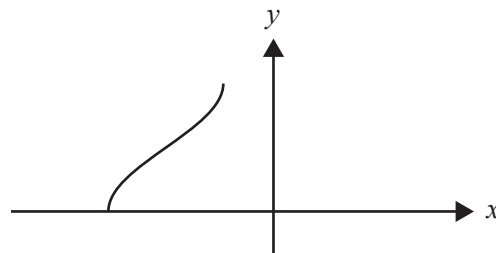
B.



C.



D.



9 What is the domain of f where $f(x) = \cos^{-1}(\ln(bx))$, where $b > 0$?

A. $\left[\frac{1}{b}, \frac{e}{b}\right]$

B. $\left[\frac{1}{b}, \frac{e^\pi}{b}\right]$

C. $\left[\frac{1}{be}, \frac{e}{b}\right]$

D. $\left[\frac{1}{be}, \frac{e^\pi}{b}\right]$

10 A given function $f(x)$ has an inverse $f^{-1}(x)$.

The derivatives of $f(x)$ and $f^{-1}(x)$ exist for all real numbers x .

The graphs $y = f(x)$ and $y = f^{-1}(x)$ have at least one point of intersection.

Which statement is true for all points of intersection of these graphs?

A. All points of intersection lie on the line $y = x$.

B. None of the points of intersection lie on the line $y = x$.

C. At no point of intersection are the tangents to the graphs parallel.

D. At no point of intersection are the tangents to the graphs perpendicular.

Section III – Part A

Label your Answer Booklet as Section III Part A

Part A – 15 marks

Attempt Questions 11 – 15

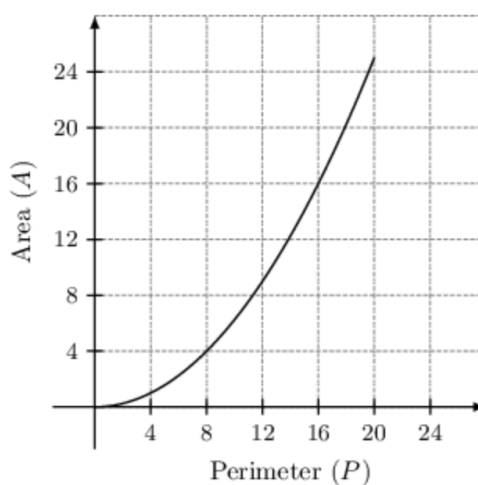
Your responses should include relevant mathematical reasoning and/or calculations.

- 11 The area, A , of a given square can be represented by the function

$$A(P) = \frac{P^2}{16}, P \geq 0,$$

where P is the perimeter of the square.

The graph of the function A , for $0 \leq P \leq 20$, is shown below.



- (a) Find the value of $A(20)$. 1
- (b) Copy the diagram to your Answer booklet, and on it draw the graph of the inverse function, A^{-1} . 2
- (c) In the context of the question, explain the meaning of $A^{-1}(4) = 8$. 1
- 12 Sketch the function $f(x) = 3\arcsin(2x)$. 2
Clearly show any intersections with the coordinate axes and endpoints.
- 13 (a) Given that $x^2 + 4x + 5 \equiv (x + a)^2 + b^2$, find a and b . 1
- (b) Hence, using the substitution $u = x + a$, find $\int \frac{1}{x^2 + 4x + 5} dx$. 2

- 14** Let $f(x) = \pi + \arccos(2x - 1) - 2\arccos(\sqrt{x})$ for $x \in [0, 1]$.
- (a) Show that $f'(x) = 0$. **2**
- (b) Hence, find a simplified form of $f(x)$. **1**
-
- 15** The function f is defined as $f(x) = \sec x$ for $0 \leq x < \frac{\pi}{2}$.
- (a) State the domain of the inverse function $f^{-1}(x)$. **1**
- (b) Show that $f^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right)$. **2**

End of Part A

Section III – Part B

Label your Answer Booklet as Section III Part B

Part B 15 marks

Attempt Questions 16 – 18

Your responses should include relevant mathematical reasoning and/or calculations.

16 Use the substitution $x = \cot \theta$ to evaluate $\int \frac{1}{(x^2 + 1)^2} dx$. 4

You may use without proof that $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

17 Use the substitution $u = x \ln x$ to evaluate $\int_1^2 \frac{\ln x^2 + 2}{(x \ln x + 1)^2} dx$. 4

18 Consider the function h : $h(x) = |\ln(2x)| + 1$, $0 < x \leq \lambda$, where $\lambda \in \mathbb{R}$.

(a) By means of a simple graph, show that the maximum value of λ for which the inverse function of h exists is $\lambda = \frac{1}{2}$. 1

(b) Let $\lambda = \frac{1}{2}$.

(i) Find $h^{-1}(x)$ and state its domain. 3

(ii) Sketch the graph of $y = (h \circ h^{-1})(x)$. 2

(iii) Find the solution set for $h(x) = h^{-1}(x)$. 1

End of paper



**SYDNEY
BOYS
HIGH
SCHOOL**

2023

YEAR 12
TERM 3
ASSESSMENT TASK 3

Mathematics Extension 1

Sample Solutions

MC Answers

1	A	6	C
2	B	7	C
3	B	8	A
4	B	9	C
5	B	10	D

Section II Multiple Choice Solutions

- 1 Let $f(x) = \operatorname{cosec} x$.

The graph of f is transformed by

- a vertical dilation by a factor of 3, followed by
- a translation of 1 unit horizontally to the right, followed by
- a horizontal dilation by a factor of $\frac{1}{2}$.

Which of the following functions is the transformed graph?

A.

$$g(x) = 3\operatorname{cosec}(2x - 1)$$

B.

$$g(x) = 3\operatorname{cosec}\left(\frac{x}{2} - 1\right)$$

C.

$$g(x) = 3\operatorname{cosec}(2(x - 1))$$

D.

$$g(x) = 3\operatorname{cosec}\left(\frac{x - 1}{2}\right)$$

$$f(2x - 1)$$

Shift 1 unit to the right followed by a dilation horizontally by a factor of $\frac{1}{2}$

This is the same as $f\left(2\left(x - \frac{1}{2}\right)\right)$, which is dilating by a factor of $\frac{1}{2}$ and then shifting to the right by a $\frac{1}{2}$ unit.

$$f(2(x - 1))$$

A dilation horizontally by a factor of $\frac{1}{2}$ followed by a shift 1 unit to the right

This is the same as $f(2x - 2)$, which is translating 2 units to the right followed by a horizontal dilation by a factor of $\frac{1}{2}$.

34 students got this question right

- 2 Which of the following is the inverse function of $f(x) = 2x^{\frac{1}{4}}$?

A. $f^{-1}(x) = \frac{x^{\frac{1}{4}}}{16}$

B. $f^{-1}(x) = \frac{x^4}{16}$

C. $f^{-1}(x) = \frac{x^4}{8}$

D. $f^{-1}(x) = \frac{x^4}{2}$

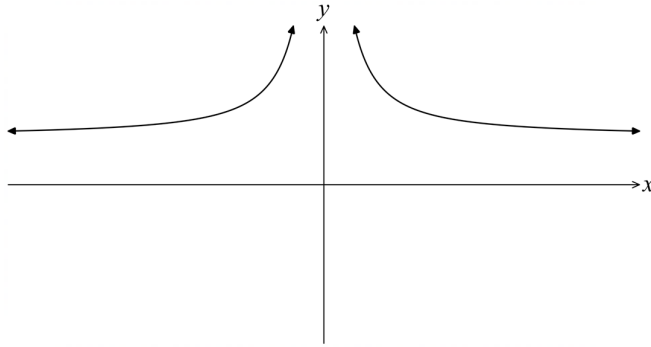
$$y = 2x^{\frac{1}{4}} \Rightarrow \frac{y}{2} = x^{\frac{1}{4}}$$

$$\therefore x = \frac{1}{16}y^4$$

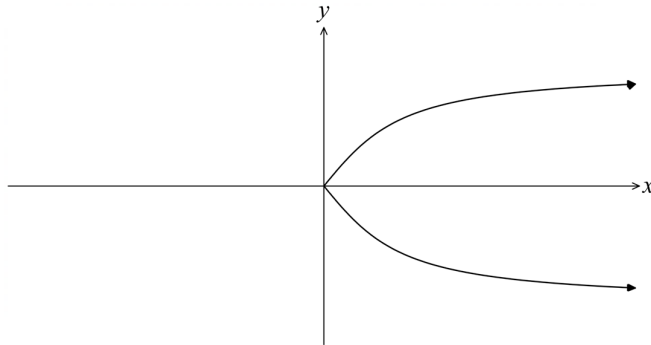
Now swap x and y .

3 Which of the following sketches best represents $|y| = \arctan x$?

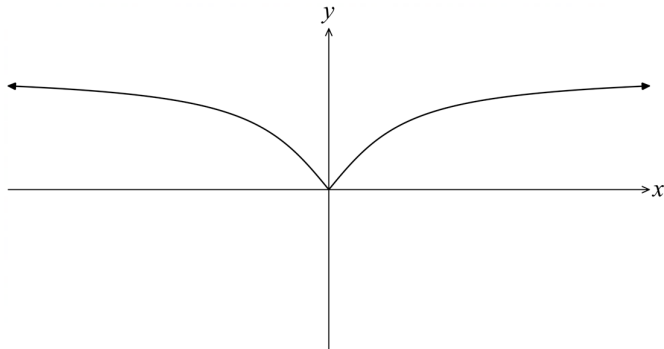
A.



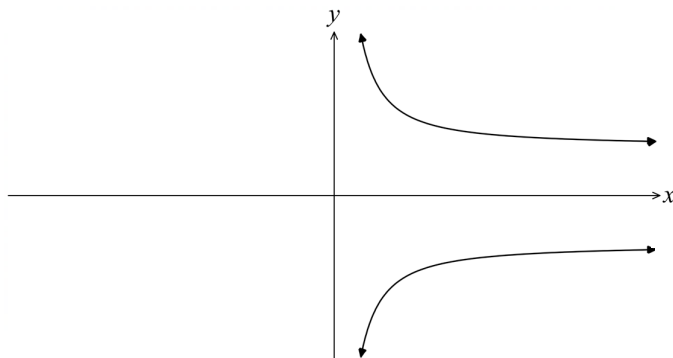
B.



C.



D.



With the symmetry about the x -axis for $|y| = f(x)$. It has to be B or D.

$f(x) = \arctan x$ has no vertical asymptote.

So B

4 Which of the following is equivalent to $\int_0^{\frac{\pi}{6}} \sin^5 x \cos x \, dx$, after applying the substitution $u = \sin x$?

A. $\int_0^{\frac{\pi}{6}} u^5 \, du$

☒ B. $\int_0^{\frac{1}{2}} u^5 \, du$

C. $-\int_0^{\frac{\pi}{6}} u^5 \, du$

D. $-\int_0^{\frac{1}{2}} u^5 \, du$

Has to be B or D with the change in upper and lower limits.

As the u -sub is $u = \sin x$ then $du = \cos x \, dx$

B

5 Given that $f(x) = e^x - 1$ and $y = f^{-1}(x)$, which of the following is an expression for $\frac{dy}{dx}$?

A. $\frac{1}{e^x - 1}$

☒ B. $\frac{1}{x + 1}$

C. $\ln x$

D. $\ln(x + 1)$

$$f(x) = e^x - 1 \Rightarrow x = \ln(1 + y)$$

$$\therefore f^{-1}(x) = \ln(1 + x)$$

Using:

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

$$\therefore \frac{dy}{dx} = \frac{1}{e^{\ln(x+1)}} = \frac{1}{x+1}$$

OR

$$f(x) = e^x - 1 \Rightarrow x = \ln(1 + y)$$

$$\therefore y = \ln(1 + x) \Rightarrow \frac{dy}{dx} = \frac{1}{1+x}$$

6 What is the value of $\arcsin(\cos \alpha)$, where $\pi < \alpha < \frac{3\pi}{2}$?

A. $\frac{\pi}{2} - \alpha$

B. $\frac{3\pi}{2} + \alpha$

C. $\alpha - \frac{3\pi}{2}$

D. $\frac{3\pi}{2} - \alpha$

As the results are linear, just test the end points π and $\frac{3\pi}{2}$.

$$\arcsin(\cos \pi) = \arcsin(-1) = -\frac{\pi}{2}$$

i.e.

$$\arcsin\left(\cos \frac{3\pi}{2}\right) = \arcsin(0) = 0$$

$\therefore$ C

OR

Now $\arcsin(\sin u) = u$ if $-\frac{\pi}{2} \leq u \leq \frac{\pi}{2}$

$$\arcsin(\cos \alpha) = \arcsin[\cos(2\pi - \alpha)]$$

$$= \arcsin\left[\sin\left(\frac{\pi}{2} - (2\pi - \alpha)\right)\right]$$

$$= \arcsin\left[\sin\left(-\frac{3\pi}{2} + \alpha\right)\right]$$

$$= -\frac{3\pi}{2} + \alpha$$

Why?

$$\pi \leq \alpha \leq \frac{3\pi}{2} \Rightarrow \pi - \frac{3\pi}{2} \leq -\frac{3\pi}{2} + \alpha \leq \frac{3\pi}{2} - \frac{3\pi}{2}$$

$$\therefore -\frac{\pi}{2} \leq -\frac{3\pi}{2} + \alpha \leq 0$$

7 If $y = \sin^{-1}\left(\frac{a}{x}\right)$ then which of the following is $\frac{dy}{dx}$, where a is a constant?

A. $-\frac{a}{x^2\sqrt{x^2-a^2}}$

B. $-\frac{a}{x^2\sqrt{a^2-x^2}}$

C. $-\frac{a}{|x|\sqrt{x^2-a^2}}$

D. $-\frac{a}{|x|\sqrt{a^2-x^2}}$

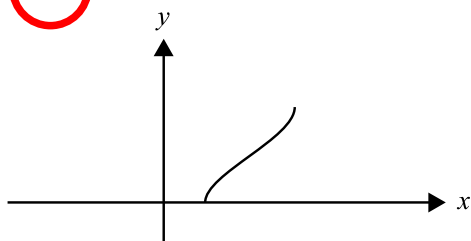
$$y = \sin^{-1}\left(\frac{a}{x}\right) \Rightarrow \frac{dy}{dx} = -\frac{a}{x^2} \times \frac{1}{\sqrt{1-\left(\frac{a}{x}\right)^2}}$$

From here the answer is A or C

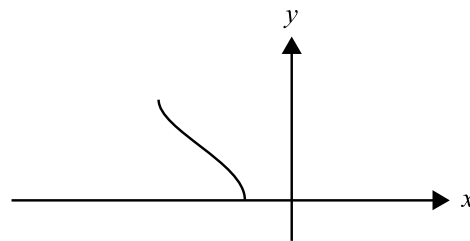
$$\frac{dy}{dx} = -\frac{a}{x^2} \times \frac{1}{\sqrt{\frac{x^2-a^2}{x^2}}} = -\frac{a}{x^2} \times \frac{|x|}{\sqrt{x^2-a^2}} = -\frac{a}{|x|\sqrt{x^2-a^2}}$$

8 Which of the following could be the graph of $y = \arccos(2 - bx)$, where $b > 0$?

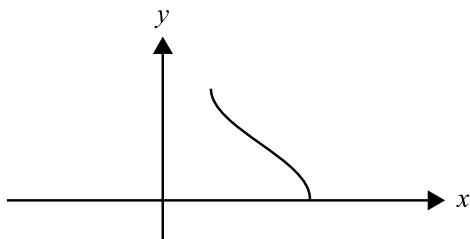
A.



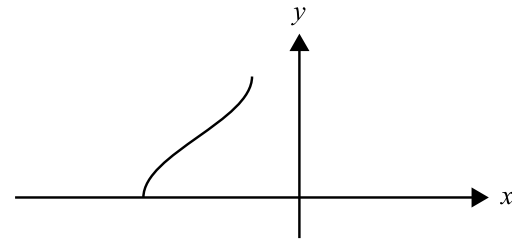
B.



C.



D.



$$f(2 - bx) = f\left(-b\left(x - \frac{2}{b}\right)\right): \text{ A reflection in the } y\text{-axis with a horizontal dilation factor of } \frac{1}{b}$$

Then a horizontal translation to the right by $\frac{2}{b}$ units

9 What is the domain of f where $f(x) = \cos^{-1}(\ln(bx))$, where $b > 0$?

A. $\left[\frac{1}{b}, \frac{e}{b}\right]$

B. $\left[\frac{1}{b}, \frac{e^\pi}{b}\right]$

☒ C. $\left[\frac{1}{be}, \frac{e}{b}\right]$

D. $\left[\frac{1}{be}, \frac{e^\pi}{b}\right]$

We need $-1 \leq \ln(bx) \leq 1 \Rightarrow e^{-1} \leq bx \leq e$

$$\therefore \frac{1}{be} \leq x \leq \frac{e}{b}$$

10 A given function $f(x)$ has an inverse $f^{-1}(x)$.

The derivatives of $f(x)$ and $f^{-1}(x)$ exist for all real numbers x .

The graphs $y = f(x)$ and $y = f^{-1}(x)$ have at least one point of intersection.

Which statement is true for all points of intersection of these graphs?

A. All points of intersection lie on the line $y = x$.

B. None of the points of intersection lie on the line $y = x$.

C. At no point of intersection are the tangents to the graphs parallel.

☒ D. At no point of intersection are the tangents to the graphs perpendicular.

A is false. Some inverse functions will intersect on $y = -x$ e.g. let $f(x) = -x$

B is patently false

C is false. Think about $f(x) = x$

D is true as the statement is false.

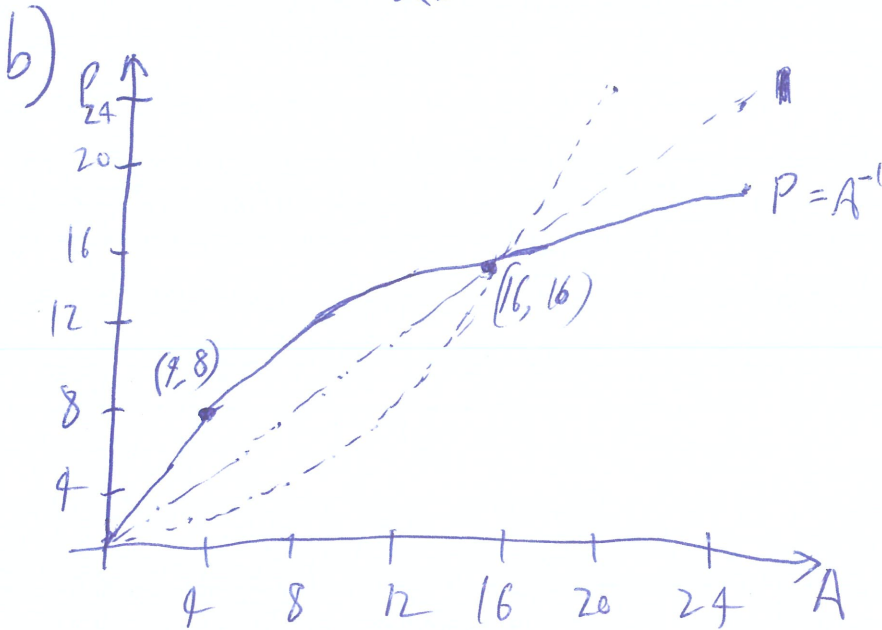
50 students got this question right

Extension 1 Task 3

①

Section III Part A

1) a) $A(20) = \frac{20^2}{16}$
 $= 25$



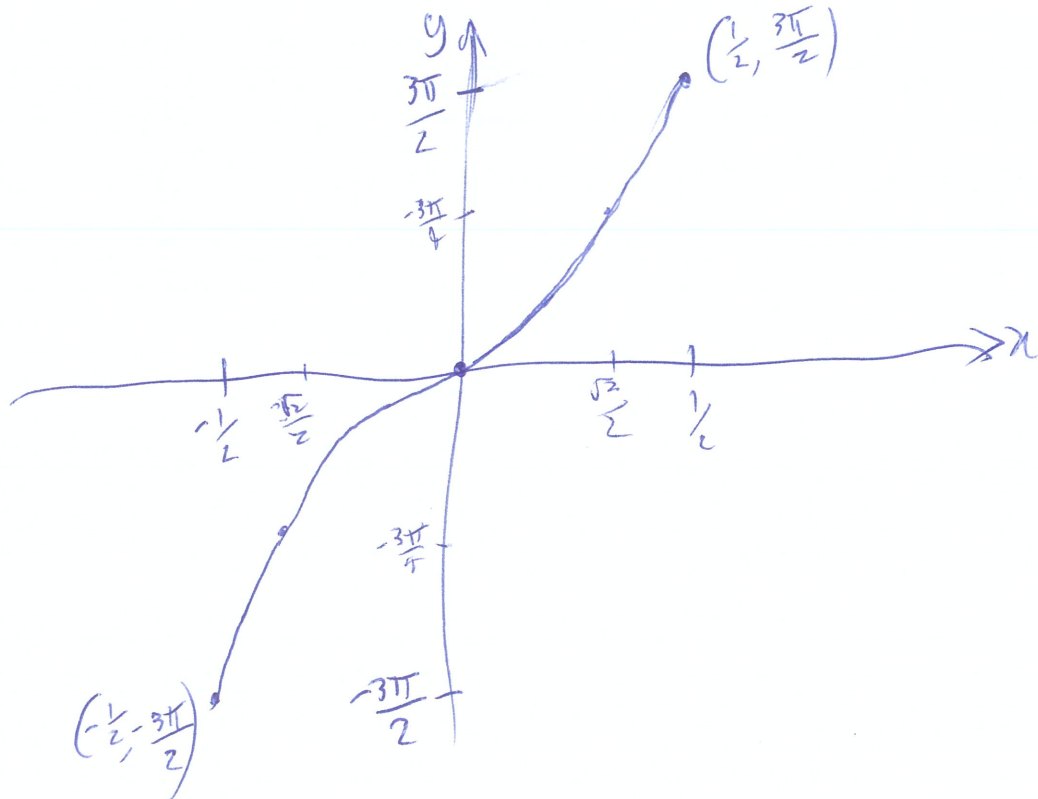
Plot some point(s) to draw through.

c) $A^{-1}(4) = 8$ means that a square of area 4 has perimeter 8.

The keyword here is "context", which indicates you need to refer to the real world application of the formula.

12.

$2x$	-1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1
x	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{4}$	0	$\frac{\sqrt{2}}{4}$	$\frac{1}{2}$
$\arcsin(2x)$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$3\arcsin(2x)$	$-\frac{3\pi}{2}$	$-\frac{3\pi}{4}$	0	$\frac{3\pi}{4}$	$\frac{3\pi}{2}$



Be careful with contracting or expanding. This was mostly done well. A table of values can help.

$$13. a) \text{ RHS} = (x+a)^2 + b^2 \\ = x^2 + 2ax + (a^2 + b^2)$$

$$\text{LHS} = x^2 + 4x + 5$$

$$\text{So } 2a = 4$$

$$a = 2.$$

Compare coefficients

$$\text{Then } a^2 + b^2 = 5$$

$$b^2 = 5 - 2^2$$

$$= \pm 1.$$

$$\therefore a = 2, b = \pm 1. \quad (\text{and } x^2 + 4x + 5 = (x+2)^2 + 1^2).$$

$$b) \int \frac{1}{x^2 + 4x + 5} dx = \int \frac{1}{u^2 + 1^2} du$$

$$= \arctan(u) + C$$

$$= \arctan(x+2) + C.$$

$$\text{Let } u = x+2 \\ du = dx$$

(from (a))

ALWAYS check how a previous part can help the current part.

Some students need to become more familiar with the standard integrals, so you can find the pattern quickly.

14a) $f(x) = \pi + \arccos(2x-1) - 2 \arccos(\sqrt{x})$ for $x \in [0, 1]$

4

$$\frac{d}{dx} (\arccos(2x-1)) = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{2}{\sqrt{1-u^2}}$$

$$= \frac{2}{\sqrt{1-(2x-1)^2}}$$

$$= \frac{2}{\sqrt{-4x^2+4x}}$$

$$= \frac{1}{\sqrt{-x^2+x}}$$

$$\frac{d}{dx} (2 \arccos(\sqrt{x})) = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{2}{2\sqrt{u}\sqrt{1-u^2}}$$

$$= \frac{1}{\sqrt{x-x^2}} \text{ as above.}$$

$$\therefore f'(x) = 0$$

Recognise the chain rule straight away.

Don't leave out terms on each line just to save writing.

b) We know $f(x) = C$ is $f'(x) = 0$ on the domain.

$$\text{Try } f(0) = \pi + \arccos(-1) - 2 \arccos(0)$$

$$= \pi + \pi - 2\left(\frac{\pi}{2}\right)$$

$$= \pi.$$

$$\therefore f(x) = \pi \text{ on the domain.}$$

Done well.

Again use part (a), and be familiar with the meaning of the derivative.

15. a) Notice that on $[0, \frac{\pi}{2})$ that $\cos x$ goes from 1 to 0.

So its reciprocal, $\sec x$ goes

from $\frac{1}{1}$ to $\lim_{x \rightarrow \frac{\pi}{2}} \frac{1}{\cos x} = \infty$.

$\therefore$ the domain is $1 \leq x < \infty$
or $[1, \infty)$.

Notice the round parenthesis on the end.

b) $f(x) = \sec x$

i.e. $y = \sec x$

Swap x and y for the inverse.

$$x = \sec y$$

$$= \frac{1}{\cos y}$$

So $\cos y = \frac{1}{x}$

Take $\cos^{-1}$ of both sides

$$y = \cos^{-1} x = f^{-1}(x).$$

16 Use the substitution $x = \cot \theta$ to evaluate $\int \frac{1}{(x^2 + 1)^2} dx$.

4

You may use without proof that $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

$$x = \cot \theta \text{ so } dx = -\operatorname{cosec}^2 \theta d\theta$$

$$I = \int \frac{1}{(x^2 + 1)^2} dx$$

$$= \int \frac{-1}{((\cot \theta)^2 + 1)^2} \cdot \operatorname{cosec}^2 \theta d\theta$$

$$= \int \frac{-1}{(\csc \theta)^2} d\theta$$

$$= -\int (\sin \theta)^2 d\theta$$

$$= \frac{1}{2} \int (\cos 2\theta - 1) d\theta$$

$$= \frac{1}{2} \left[\frac{1}{2} \sin 2\theta - \theta \right] + C$$

$$= \frac{1}{2} \left[\frac{\cot \theta}{(\cot \theta)^2 + 1} - \theta \right] + C$$

$$= \frac{1}{2} \left[\frac{x}{x^2 + 1} - \cot^{-1} x \right] + C +$$

Many candidates failed to correctly differentiate the given substitution.

One mark was lost for not putting the answer back in terms of x .

Half a mark was lost for not simplifying the re-substitution.

17 Use the substitution $u = x \ln x$ to evaluate $\int_1^2 \frac{\ln x^2 + 2}{(x \ln x + 1)^2} dx$.

4

$$u = x \ln x \text{ so } du = \left(x \cdot \frac{1}{x} + \ln x \right) dx \quad \text{when } x = 1, u = 0, \text{ and when } x = 2, u = 2 \ln 2$$

$$= (1 + \ln x) dx$$

$$I = \int_1^2 \frac{\ln(x^2) + 2}{(x \ln x + 1)^2} dx = \int_1^2 \frac{2(\ln x + 1)}{(x \ln x + 1)^2} dx$$

$$= \int_0^{2 \ln 2} \frac{2}{(u+1)^2} du$$

$$= 2 \int_0^{2 \ln 2} (u+1)^{-2} du$$

$$= 2 \left[\frac{-1}{u+1} \right]_0^{2 \ln 2}$$

$$= 2 \left[1 - \frac{1}{1+2 \ln 2} \right]$$

$$= \frac{4 \ln 2}{1+2 \ln 2} \approx 1.16188$$

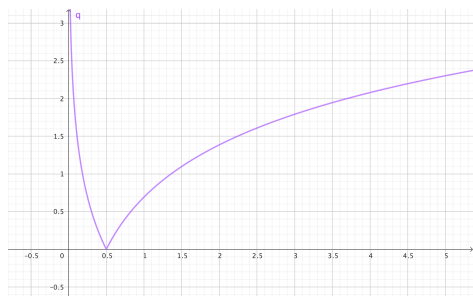
Many candidates failed to correctly differentiate the given substitution.
Otherwise this was generally well answered.

Several candidates read $\ln x^2$ as $(\ln x)^2$; they were given marks for any correct working even though it was leading nowhere.]

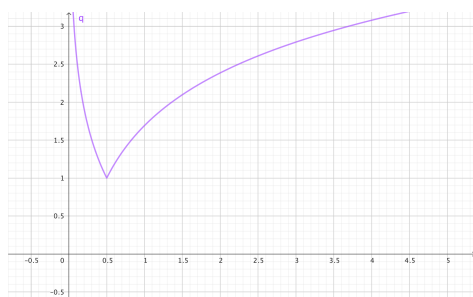
18 Consider the function h : $h(x) = |\ln(2x)| + 1, 0 < x \leq \lambda$, where $\lambda \in \mathbb{R}$.

- (a) By means of a simple graph, show that the maximum value of λ for which the inverse function of h to exist is $\lambda = \frac{1}{2}$.

1



$$y = |\ln(2x)|$$



$$y = |\ln(2x)| + 1$$

$$\text{Domain } \{x: x > 0\}$$

$$\text{Range } \{y: y \geq 1\}$$

Clearly $h(x)$, with domain $\{x: 0 < x \leq \lambda\}$ is one-to-one only until $\lambda = \frac{1}{2}$ (horizontal line test), and hence may be inverted to a function in this domain.

Part marks were lost for not mentioning either one-to-one or the horizontal line test.

- (b) Let $\lambda = \frac{1}{2}$.

- (i) Find $h^{-1}(x)$ and state its domain.

3

$$\text{Swap } x \text{ and } y: x = |\ln(2y)| + 1$$

$$x - 1 = \ln(2y)$$

or

$$1 - x = \ln(2y)$$

$$y = \frac{e^{x-1}}{2}$$

or

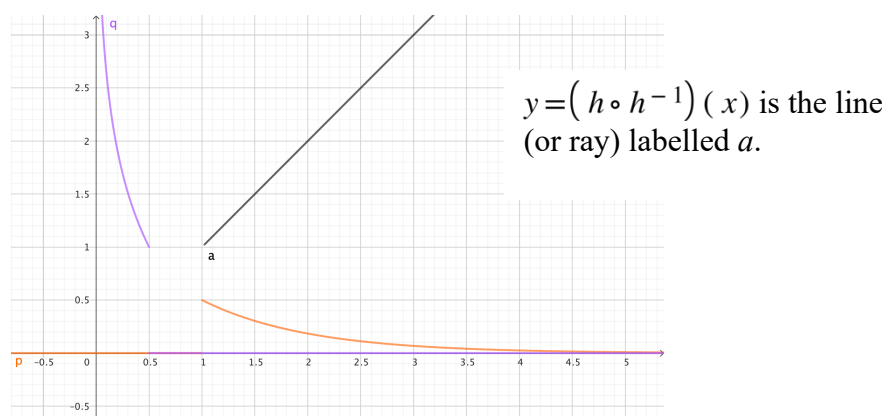
$$y = \frac{e^{1-x}}{2}$$

Range is $\{y: 0 < y \leq \frac{1}{2}\}$, so the inverse is $y = \frac{e^{1-x}}{2}$, with Domain $\{x: x \geq 1\}$

Many candidates chose the wrong branch, or the wrong domain, and so did not gain marks.

- (b) (ii) Sketch the graph of $y = (h \circ h^{-1})(x)$.

2



One mark was awarded for the whole line $y = x$, and for the graph $y = |x| + 1$.

The diagonal from $(0, 0)$ to $(1, 1)$ received no marks as the domain and range do not match their answer.

- (iii) Find the solution set for $h(x) = h^{-1}(x)$.

1

The solution set is ϕ , the empty set (by inspection).

Many candidates failed to notice that the domains do not overlap.

Overall, many candidates need to make a greater effort at presentation of their answers.

The marker should not have to search amongst illegible scrawling to find something to award a mark for.

Similarly several candidates need to make their written characters distinct. For instance a “4” should not look like “u”, and such like.

End of solutions